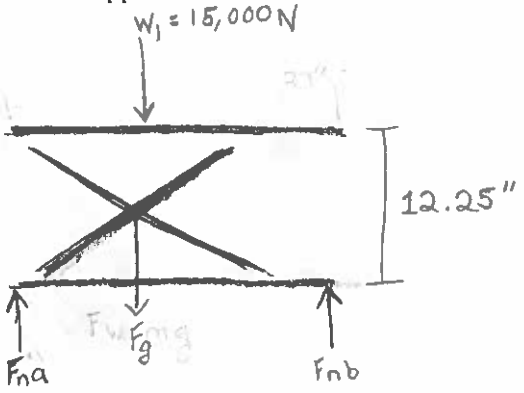
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Requirements 01: (4 marks)

A FBD sketch should include:

- **A simplified outline:** Sketch a clean, basic shape of the outer boundaries. Leave out all internal details.
- **Completely isolated:** Do not draw the walls, floors, pins, or hinges that it contacts.
- **External forces:** Include all external force vectors, starting at the exact locations where they actually push or pull on the outline, as well as moments and internal forces such as gravity.

Insert the FBD of the scissor lift at its lower height, including only the scissor bars, top platform, and lower platform. Show all applied forces and determine all reaction forces.



Lowest Height:
22" - 9.75" = 12.25"

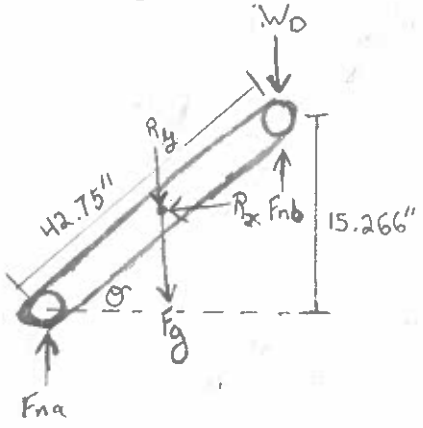
Normal Force A = 7,500

Normal Force B = 7,500

Insert the FBD of the scissor lift bar (one of the scissor sides) when the top platform is at its maximum height.

The FBD must:

- Show all applied forces and reaction forces acting on the bar.
- Determine all reaction forces.
- Clearly show the angle of the scissor bar relative to the horizontal (x-axis).



$\theta = \sin^{-1}\left(\frac{Opp}{Hyp}\right)$

$= \sin^{-1}\left(\frac{15.266}{42.75}\right)$

$= 20.9°$

$F_g = mg$

$F_{na} = F_g/2$

$F_{nb} = F_g/2$

R_y - Reaction Force y

R_x - Reaction Force x

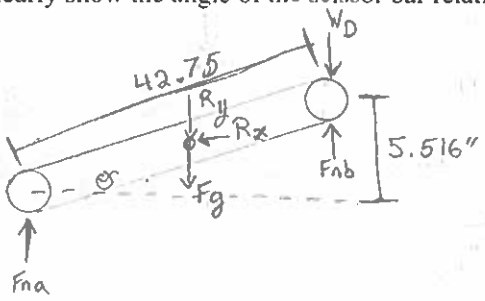
W_D - Weight Distributed

$W_D = 7,500\ N$

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Insert the FBD of the scissor lift bar when the top platform is at its lowest height.
The FBD must:

- Show all applied forces and reaction forces acting on the bar.
- Determine all reaction forces.
- Clearly show the angle of the scissor bar relative to the horizontal (x-axis).



Lowest Height:
$15.266" - 9.75" = 5.516"$

$\theta = \sin^{-1}\left(\frac{Opp}{Hyp}\right)$
$\theta = \sin^{-1}\left(\frac{5.516}{42.75}\right)$
$\theta = 7.4°$

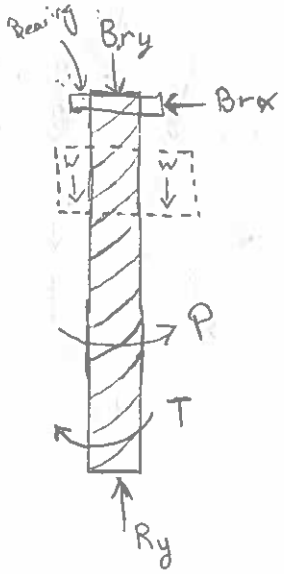
R_y - Reaction Force y
R_x - Reaction Force x

$F_g = mg$
$F_{na} = F_g/2$
$F_{nb} = F_g/2$

W_D - Weight Distributed
$W_D = 7,500N$

Insert the FBD of the Acme screw when the lift any position".
The FBD must:

- Show all forces and moments acting on the Acme screw.
- Identify the applied forces and reaction forces.
- Determine all reaction forces and moments.



T → Torque
Ry → Reaction Force y
W → Weight
Bry → Bearing reaction y
Brx → Bearing reaction x

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Requirements 02: (3 marks)

Calculate to design, verify and confirm the acme screw used for this design.

Investigate options to find;

- the friction angle ✓
- the helix angle ✓
- P; Effort applied at the circumference of the screw to lift the load of 15000N ✓
- maximum required torque, ✓
- maximum required power ✓
- calculate the efficiency. ✓
- Verify the selected gear motor in the design attached to this assignment (use Oriental motors website)

ACME:
$2\beta = 29°$
$\beta = 14.5$

Trap:
$2\beta = 30°$
$\beta = 15°$

Given:

$W = 15\,000\ N$

Stroke = 9.75" → 248 mm

Time = 90sec

Velocity = 2.756 mm/s → $\frac{2.756}{1000}$ → 0.002756 m/s

Outlet:

V = 110 VAC

I = 15 Amp

McMaster-Carr Part #:
7549K81

Lead = 5mm

Pitch = 5mm

diameter = 24mm

$h = \frac{P}{2}$

$h = \frac{5mm}{2}$

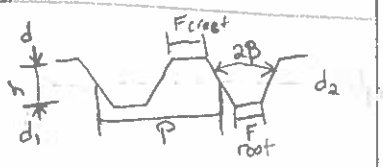
$h = 2.5mm$

Helix Angle:

$\tan\lambda = \frac{L}{\pi d_2}$

$\lambda = \tan^{-1}\left(\frac{5mm}{\pi\, 21.5mm}\right)$

$\lambda = 4.23°$



d = diameter
h = height
d_1 = Minor diameter of external thread
d_2 = Pitch diameter of external thread
P = Pitch
F = Flat width

$d_2 = d - h$

$d_2 = 24mm - 2.5mm$

$d_2 = 21.5mm$

Friction Angle:

$\phi = \tan^{-1}\left(\frac{0.15}{\cos(15)}\right)$

$\phi = 8.83°$

Using $\mu = 0.15$
Hardened steel on Cast Iron

Calculating P:

$\tan\lambda = \frac{p}{\pi d_a}$

$\tan\lambda = \frac{5mm}{\pi \cdot 21.6mm}$

$\tan\lambda = 0.074$

$\tan\phi = \left(\frac{\mu}{\cos\beta}\right)$

$\tan\phi = \left(\frac{0.15}{\cos(15)}\right)$

$\tan\phi = 0.1553$

$P = W\tan(\lambda+\phi) = W\left[\frac{\tan\lambda + \tan\phi}{1-\tan\lambda\tan\phi}\right]$

$P = 15\times10^3\left[\frac{0.074+0.1553}{1-0.074\cdot0.1553}\right]$

$P = 3479.5\ N$

Max Torque:

$T = P\cdot\frac{d}{2}$

$T = 3479.5N\cdot\frac{24mm}{2}$

$T = 41754\ N\text{-}mm$

$\downarrow \times 0.001$

$T = 41.754\ N\text{-}m$

Efficiency:

$P_o = W\tan\lambda$

$P_o = 15\times10^3\cdot0.074$

$P_o = 1110N$

$\eta = \frac{P_o}{P}$

$\eta = \frac{1110N}{3479.5N}$

$\eta = 0.32\cdot100$

$\eta = 32\%$

Max Power:

$N = \frac{Velocity}{P\cdot tan}$

$N = \frac{0.002756\ m/s}{0.005m}$

$N = 0.6512\ rev/s\cdot60$

$N = 33.1\ rpm$

$w = \frac{2\pi N}{60}$

$w = \frac{2\pi(33.1)}{60}$

$w = 3.5\ rads/s$

Power $= T\cdot w$

Power $= 41.754Nm\cdot 3.5\ rad/s$

Power $= 146.14\ W$

Selected Gear Motor is Good